

COURSE CERTIFIED BY:
COURSE CERTIFIED THRU:

Mission Systems Written Comp Review

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United States Air Force **Test Pilot School**



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Topics

- Module 1: Foundational Principles
- Module 2: Radar Systems
- Module 3: Electro-Optic (EO) & Infrared (IR) Systems
- Module 4: Navigation & Guidance
- Module 5: Electronic Warfare
- Module 6: Communications & Datalinks
- Module 7: Munitions
- Module 8: Key Supporting Concepts (SWaP-C, AI, Cyber, etc.)

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MODULE 1: FOUNDATIONAL PRINCIPLES

United States Air Force **Test Pilot School**





MODULE 1: FOUNDATIONAL PRINCIPLES

- **Core Concept: The Electromagnetic (EM) Spectrum**

- The EM spectrum is the entire range of all types of electromagnetic radiation. This energy travels in waves and is differentiated by its wavelength (λ) and frequency (f). These two properties are inversely proportional, linked by the speed of light (c , a constant $\approx 3 \times 10^8$ m/s).
 - Equation: $c = f * \lambda$
- This means a long wavelength corresponds to a low frequency, and a short wavelength corresponds to a high frequency. To find the wavelength of a signal given its frequency, you can rearrange this to $\lambda = c / f$.

- **Application:**

- Understanding the relative position of different systems on the EM spectrum is crucial. Generally, the order from longest wavelength to shortest is:
Radio > Microwave > Infrared > Visible Light > Ultraviolet > X-ray > Gamma Ray.



MODULE 2: RADAR SYSTEMS



MODULE 2: RADAR SYSTEMS

- **Core Concept: The Radar Range Equation & Key Parameters**

- A radar's ability to detect a target is governed by the radar range equation. Its performance hinges on several key factors:
 - Antenna Gain (G): A measure of the antenna's ability to focus power. Gain is often expressed in decibels relative to an isotropic radiator (dBi). You can calculate it from the antenna's directive gain ratio.
 - Equation: $G(\text{dBi}) = 10 * \log_{10}(\text{Gain Ratio})$
 - Radar Cross Section (RCS, or σ): A measure of how "big" a target appears to a radar. A key concept is that 0 dBsm is not "invisible"; it simply means the target has the RCS of a perfect one-square-meter sphere. In the radar range equation, detection range is proportional to the fourth root of RCS ($R \propto \sqrt[4]{\sigma}$). This leads to a crucial relationship for calculating changes in RCS.
 - Equation: $\text{RCS}_{\text{new}} = \text{RCS}_{\text{old}} * (\text{Range}_{\text{new}} / \text{Range}_{\text{old}})^4$
 - Signal-to-Noise Ratio (SNR): The ratio of the desired signal (the echo) to the background noise. A negative SNR (e.g., -8 dB) does not guarantee safety or prevent detection; modern radars use pulse integration (processing gain) to "pull" weak signals out of the noise floor.



MODULE 2: RADAR SYSTEMS

- **Application:**

- RCS Reduction: The primary goal of stealth technology. The hierarchy of design principles is: 1. Shaping, 2. Materials, and 3. Size.
- Doppler Clutter: Radars use the Doppler effect to distinguish moving targets from stationary ground clutter. On a Doppler spectrum display, returns with a negative frequency shift (to the left of the main lobe) are from objects receding from the radar.
- Synthetic Aperture Radar (SAR): A powerful imaging technique. The angular width (resolution) of a SAR beam (β , in radians) is inversely proportional to the length of the synthetic aperture (L_{eff}). A smaller beamwidth means better resolution.
 - Equation: $\beta \approx 0.44 * (\lambda / L_{\text{eff}}) \rightarrow$ Where λ is the wavelength of the radar.
- Beam Spot Size: The physical width of a radar beam increases with distance. The size of the beam at a given range can be calculated using the small angle approximation.
 - Equation: Beam Width = Range * Beam Angle (in radians)



MODULE 3: EO/IR SYSTEMS



Module 3: Electro-Optic (EO) & Infrared (IR) Systems

- **Core Concept: EO/IR Imaging Fundamentals**
 - EO/IR systems create images by detecting photons. Key optical concepts include:
 - Focal Length (f): The distance from the lens to the sensor.
 - Depth of Field: The zone of acceptable sharpness in front of and behind the subject on which the lens is focused.



Module 3: Electro-Optic (EO) & Infrared (IR) Systems

- **Application: Calculating System Resolution**

- An imaging sensor is made of a grid of pixels. The Instantaneous Field of View (IFOV) is the angle covered by a single pixel and determines the sensor's resolution. The physical size of a pixel on the detector (X_d) is related to the focal length (f) and the IFOV.
 - Equation: $X_d \approx f * \tan(\text{IFOV})$
 - (For very small angles, this is often simplified to $X_d \approx f * \text{IFOV}_{\text{rad}}$)
- The IFOV itself can be found by relating the size of what's being imaged on the ground to the altitude.
 - Equation: $\text{IFOV (in radians)} \approx (\text{Ground Footprint of one pixel}) / \text{Altitude}$



MODULE 4: NAVIGATION & GUIDANCE



Module 4: Navigation and Guidance

- **Core Concept: GPS and Inertial Navigation Systems (INS)**
 - GPS/INS Integration: A Kalman filter blends data from both systems. The GPS provides accurate, periodic updates to correct the INS's drift. The INS provides continuous data between GPS updates. When GPS is available and constantly updating the INS, long-duration tests are not required to characterize the system's performance.
- **Application: Operating in Degraded Environments**
 - GPS-Denied: When GPS is jammed, the system must rely solely on the INS.
 - GPS-Spoofed: This is more dangerous than jamming. A resilient system must have internal filters to compare the GPS data with other sources to identify and reject the corrupted data.



MODULE 5: ELECTRONIC WARFARE



Module 5: Electronic Warfare

- **Core Concept: Electronic Warfare (EW)**
 - EW is the battle for control of the electromagnetic spectrum, divided into Electronic Attack (EA), Electronic Protection (EP), and Electronic Support (ES). Techniques like "spread spectrum" are a form of EP.



MODULE 6: COMMUNICATIONS & DATALINKS



Module 6: Communications & Datalinks

- **Core Concept: Communications & Datalinks**
 - Polarization: If a receiver antenna has a polarization that is mismatched with the incoming wave (e.g., opposite circular polarizations), a massive signal loss (e.g., -25 dB) will occur.
 - One-Way vs. Two-Way Propagation: The power of a signal decreases significantly over distance, but *how* it decreases depends on whether it's a one-way or two-way link.



Module 6: Communications & Datalinks

- **Core Concept: Datalinks and Communication**

- One-Way Link (Datalinks, GPS, Jamming): Governed by the Friis Transmission Equation. Power travels from a transmitter to a receiver. Received power is proportional to $1/R^2$. This is why a jammer has a natural advantage over a radar; its signal only has to travel one way.
- Conceptual Friis Equation:
$$\text{Power_received} = (\text{Power_transmitted} * \text{Gain_transmitter} * \text{Gain_receiver} * \lambda^2) / ((4\pi R)^2)$$
- Two-Way Link (Radar): Governed by the Radar Range Equation. Power must travel to the target (a $1/R^2$ loss) and then the reflected echo must travel back to the receiver (another $1/R^2$ loss). This results in the received power being proportional to $1/R^4$.
- Conceptual Radar Range Equation: $\text{Power_received} = (\text{Power_transmitted} * G^2 * \lambda^2 * \sigma) / ((4\pi)^3 * R^4) \rightarrow$ (Where σ is the target's Radar Cross Section)
- Latency: The time delay in a communication signal. Beyond-line-of-sight satellite communications have significant latency due to the vast distances involved, making them unsuitable for time-critical actions like landing a remotely piloted aircraft.



MODULE 7: MUNITIONS



Module 7: Munitions

- **Core Concept: Weapon-Platform Communication**
 - A guided munition and its launch platform (like an aircraft) are in constant communication. This dialogue follows a distinct lifecycle, from the moment the weapon is powered on to the moment of impact. These messages ensure the weapon is healthy, correctly targeted, and safely employed. The communication can be broken down into key phases:
 - Initialization & Status Check
 - The Pre-Launch “Handshake”



Module 7: Munitions

- **Application: Key Weapon Messages**
 - Initialization & Status Check
 - Built-In Test (BIT) Messages: When a weapon is first selected by the operator, it runs a BIT to check the health of its own subsystems (guidance computer, control fins, seeker head, etc.). It then sends a "BIT Pass" or "BIT Fail" message back to the platform. A pilot cannot proceed with employment if the weapon reports a failure.
 - Alignment Status: For GPS/INS-guided weapons, the munition must align its own inertial navigation system with the platform's. It will continuously report its alignment status (e.g., "Aligned", "Degraded", "Aligning") to the operator. A successful launch requires a full alignment.



Module 7: Munitions

- **Application: Key Weapon Messages (continued)**
 - The Pre-Launch "Handshake"
 - This is the critical final sequence just before release. It's a formal, required dialogue to ensure both the weapon and the operator are ready.
 - Master Arm: The operator physically arms the aircraft's master arm switch. This is a top-level safety gate that allows power to flow for the final release sequence.
 - Data Transfer: The platform sends the final, refined targeting information to the weapon. This includes target coordinates, elevation, and impact parameters.



Module 7: Munitions

- **Application: Munition Markings (Visual Communication):**
 - Munitions use a standardized color-coding system to quickly identify their type and status to ground crews and pilots. A weapon can have multiple bands, with each color indicating the status of a different component.
 - Yellow: High Explosive (Indicates a live warhead)
 - Brown: Live Rocket Motor or Low Explosive Filler (Indicates a live propulsion system)
 - Blue: Inert (Indicates a training component, such as a warhead or motor)



MODULE 7: Munitions

Completely inert



Live Warhead (yellow) and live rocket motor (brown)



MODULE 8: KEY SUPPORTING CONCEPTS



Module 8: Key Supporting Concepts

- **Core Concept: System Development and Evaluation**
 - SWaP-C: An acronym for Size, Weight, Power, and Cost—the fundamental constraints in engineering.
 - Tracking vs. Fusion: Tracking uses a single sensor over time. Fusion combines data from multiple sensors.
 - Machine Learning: "Deep learning" is a type of machine learning that implies the use of artificial neural networks with multiple hidden layers.



Module 8: Key Supporting Concepts

- **Application: Testing and Software Development**
 - System-of-Systems Testing: Testing multiple systems together "at scale" is challenging due to cost, logistics, and data management.
 - Developmental Test (DT) Campaign: Follows a "crawl, walk, run" progression: 1. Subsystem lab tests -> 2. Software in a SIL -> 3. Hardware-in-the-Loop (HITL) -> 4. Ground tests -> 5. Flight tests.
 - Software Development Models: Waterfall (rigid, predictable) and Agile (flexible, iterative) represent a classic trade-off between predictability and responsiveness to change.



Relationships and Constants Provided

6076 feet = 1 nautical mile

1 inch = 2.54 centimeters

1 Knot = 1 nautical mile/hr = 0.5148 meters/second = 1.689 feet/sec

Speed of light = 3×10^8 meters/second

Plank's constant = 6.625×10^{-34} Joule-sec

Boltzman's constant = 1.38×10^{-23} Joule/°K

X-band Radar Doppler Rule of Thumb = 35Hz/knot

1 Parsec (PC) = 3.0856805×10^{16} meters

1 Astronomical Unit (AU) = 92.956×10^6 statute miles = 149.594×10^9 meters



Glossary of Key Acronyms

- AGL: Above Ground Level
- BIT: Built-In Test
- dBi: Decibels relative to Isotropic
- dBsm: Decibels relative to one Square Meter
- DT: Developmental Test
- EA: Electronic Attack
- EM: Electromagnetic
- EO: Electro-Optic
- EP: Electronic Protection
- ES: Electronic Support
- EW: Electronic Warfare
- FCR: Fire Control Radar
- GPS: Global Positioning System
- HITL: Hardware-in-the-Loop
- IFOV: Instantaneous Field of View
- INS: Inertial Navigation System
- IR: Infrared
- LWIR: Long-Wave Infrared
- RAM: Radar-Absorbent Material
- RCS: Radar Cross Section
- SAR: Synthetic Aperture Radar
- SIL: System Integration Lab
- SNR: Signal-to-Noise Ratio
- SOJ: Stand-Off Jammer
- SWaP-C: Size, Weight, Power, and Cost
- UHF: Ultra-High Frequency



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